

Question Paper

Exam Date & Time: 23-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026
SUBJECT: CSE 1071/CSE_1071 PROBLEM SOLVING USING COMPUTERS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Write the basic structure of a C program. Design a flowchart to count the number of divisors, other than 1 and the number itself, for a given positive integer number N (E.g. for number 6 the divisors are 1, 2, 3 and 6. So the number of divisors is 2, excluding 1 and 6). (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Evaluate the following expressions. (3)
- i) $A = 2 * 3 * 4 / (5 + 1) - 2 / 3 * (4 + 5)$
ii) $B = 2 * 3 / 4 + 4 / 4 + 8 - 2 + 5 / 8$
iii) $x = 2 > 5 != 1 ? 5 < 8 \&\& 8 > 2 ? !5 ? 10 : 20 : 30 : 40;$

Course Outcome: "CO2" / Cognitive Level: "Application"

- 1C) Develop a C program to check if the user input is a character/digit/special character using "if else" statement and if the user input is a character, then use "switch" statement to check whether the character is a vowel or a consonant and display the ASCII value of every user input. (4)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2A) Develop a C program that prints the pattern as shown below based on the given number of rows. (2)
- Sample Input: Number of rows = 6
- Output:

```
*****
*****
****
***
**
*
```

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2B) Write a C program to find the maximum product of two elements in an array of n integers. (4)
- Sample Input/output:**
Enter the number of elements: 5
Enter 5 integers: 4 3 7 9 2
The maximum product of two elements is: 63

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2C) Design a C program to find the sum of principal and secondary diagonals of a square matrix. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3A) Write a C function, multiply, which takes two integers as parameters and returns their product. Write the function prototype and mention the formal parameters. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3B) A positive integer is entered through the keyboard. Design a C function to find the binary equivalent of this number. Call this function in main() and print the results for a user-defined input. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3C) Develop a recursive C function **isPalindrome(char str[], int start, int end)** to determine whether a given string is a palindrome or not, without using any loops. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4A) A programmer writes a function `updatePrice(float price)` to increase the price of a product by 10%. However, after calling the function, the price in `main()` remains unchanged. (3)
- i) Apply your understanding of parameter passing methods to explain why the value did not change.
 - ii) Modify the function so that the price is correctly updated in the calling function.
 - iii) Write the correct C program with emphasis on the function call and function definition

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4B) Develop a C program that accepts **n integers** from the user and stores them in an array. The program should then modify the array such that: (4)

- Each **negative number** is replaced by its **square**.
- Each **positive number** is replaced by **double its value**.

Finally, display the **modified array** with appropriate messages.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) Define a structure `Vehicle` with fields: `reg_no`, `model`, `distance_travelled`, `fuel_used`, and `mileage`. Write a C program to calculate the mileage of each vehicle as `mileage = distance_travelled / fuel_used`, and display the top 3 vehicles with the highest mileage. (3)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

- 5A) Write a C program using an array of pointers to strings to store names of n students and display all names that begin with a vowel (A, E, I, O, U). (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5B) A C programmer is experimenting with pointers to understand how variables can be accessed and modified indirectly. The following declarations are made: (4)

```
int a = 10, *p;  
p = &a;
```

The pointer `p` holds the address of variable `a`, and the programmer wishes to modify `a` using the pointer variable.

Construct a complete C program to demonstrate how the dereferencing operator (*) is used to change the value of a variable through its pointer. Additionally, explain the memory-level behavior when the statement `*p = 20;` is executed, highlighting how the value stored at the address of `a` is updated.

Course Outcome: "CO5" / Cognitive Level: "Application"

5C) Compare and Contrast Surface, Deep and Dark web with an example for each. (3)

Course Outcome: "CO6" / Cognitive Level: "Understand"

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Event Name		Subject	
058-MIT BLR-MAKEUP EXAM-2025-2026		CSE_1071:PROBLEM SOLVING USING COMPUTERS	
#	Course Outcome	Description	Marks
1	1	CO 1	3
2	2	CO 2	9
3	3	CO 3	8
4	4	CO 4	17
5	5	CO 5	10
6	6	CO 6	3
Total			50

* **NA** - CO Not Selected

Event Name		Subject	
058-MIT BLR-MAKEUP EXAM-2025-2026		CSE_1071:PROBLEM SOLVING USING COMPUTERS	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	3(6%)
3	Application	12	41(82%)
4	Analysis	1	3(6%)
5	Create	1	3(6%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)